MultiPie Manual

https://cmt-mu.github.io/MultiPie/

for Version 2.2 (2026.7)

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MultiPie - Overview

Features

- **Group-theoretical data**

content	point group	space group	magnetic PG	magnetic SG
symmetry operation	✓	✓	✓	✓
Wyckoff (site)	✓	✓	✓	✓
Wyckoff (bond)	✓	✓		
character	✓			
harmonics	✓			
active multipole			✓	

- **Symmetry Adapted Multipole Basis (SAMB) data**

content	point group	space group	magnetic PG	magnetic SG
PG Clebsch-Gordan coefficinent	✓			
atomic SAMB (X)	✓			
site/bond-cluster SAMB (Y)	✓	✓		
combined SAMB (Z)	✓	✓		

- **List of group-theoretical data (PDF)**

  https://cmt-mu.github.io/MultiPie/src/group_doc.html

- **Model construction and physical quantity calculations using SAMB**

  `$ mp_create ./input_file.py` Generate SAMB from input model file

  `$ mp_analyze ./control_file.py` Compute physical quantities from control file

MultiPie - Symmetry Adapted Multipole Basis (SAMB)

Physical degrees of freedom (classified by the eigen values of the following basic symmetry operations)

- Time-reversal parity (electric or magnetic)
- Spatial-inversion property (polar or axial)
- Spatial anisotropy (rank and component)

$$X_{lm}\ (X=Q,G,T,M)$$

Time/Space	polar	axial
even	Q (electric)	G (electric toroidal)
odd	T (magnetic toroidal)	M (magnetic)

Point groups are subgroups of the rotation group

Unitary trans.

$$X_{lm} \quad \to \quad X_{l\Gamma n\gamma} = \sum_m c^{lm}_{\Gamma n\gamma} X_{lm}$$

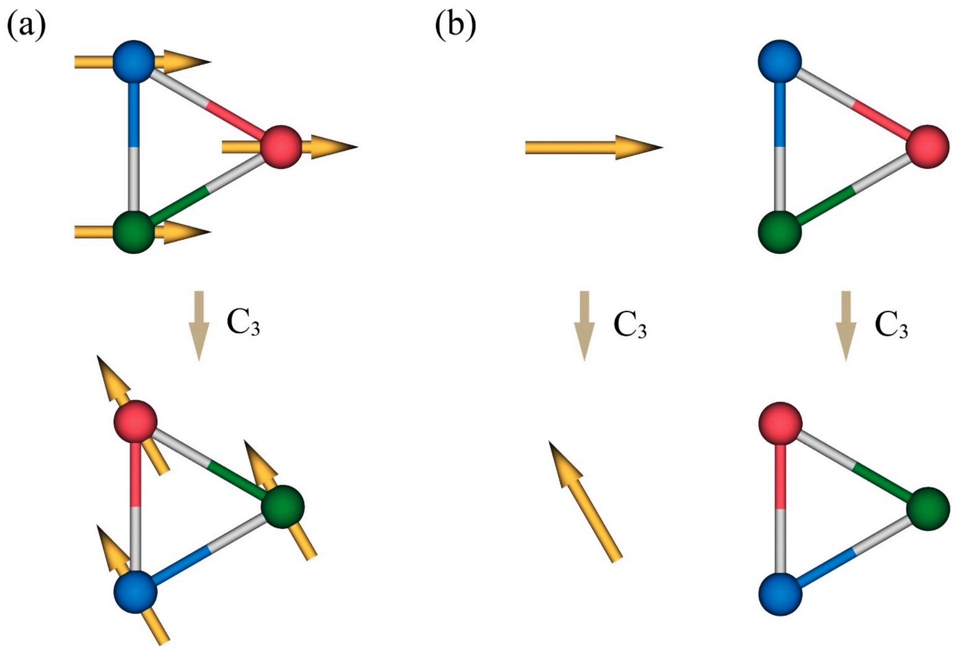
classified by PG irreps.
(Γ, n): irrep. and multiplicity
γ: component

PG symmetry op. acts independently to internal d.o.f. and structure

- internal d.o.f. (atomic multipole) X_{lm}
- structure (cluster multipole) Y_{lm}
- combined basis (combined multipole)

$$Z_{lm} = \sum_{l_1m_1,l_2m_2} c^{l_1m_1,l_2m_2}_{lm} X_{l_1m_1} Y_{l_2m_2}$$

$$\Rightarrow \{ Z_j \}$$



symmetry operation

internal d.o.f. (atomic)

structure (cluster)

MultiPie - Model Construction (Example : Graphene)

Input file

```
# graphene_in.py
graphene_in = {
    "model": "graphene",  #  name of model.
    "group": 191,  # No. of space group.
    "cell": {"c": 4},  # set large enough interlayer distance.
    #
    "site": {"C": ("[1/3,2/3,0]", "pz")},  # positions of C site and its orbital.
    "bond": [("C", "C", [1, 2])],  # C-C bonds up to 2nd neighbors.
    #
    "spinful": False,  # spinless.
    #
    "SAMB_select": {"X": []},
}
```

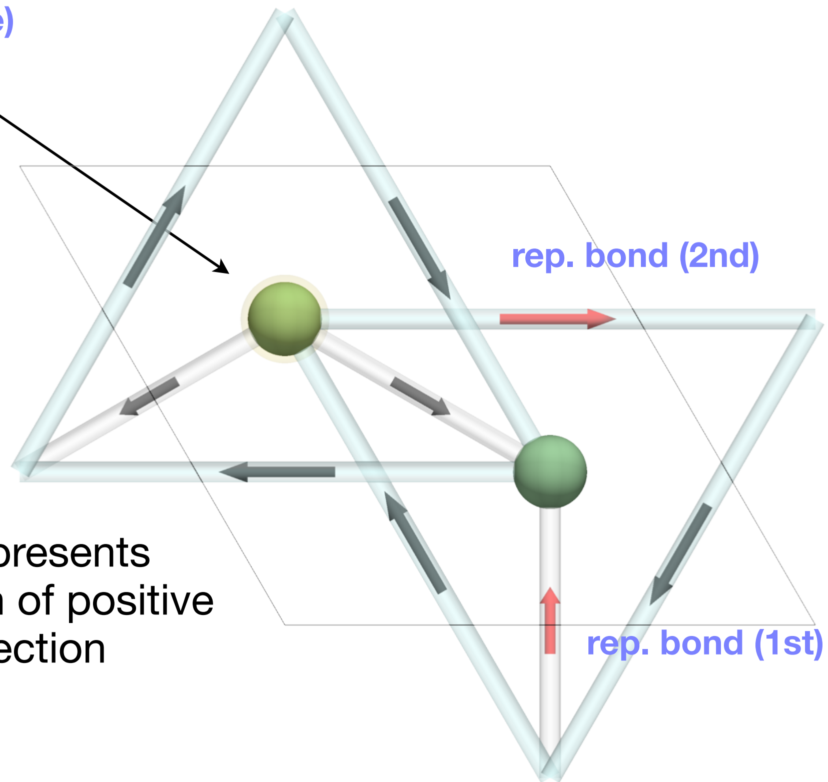
C site (representative)
p_z orbital only

C-C hopping
up to 1st and second neighbors

spinless basis

Q, G, T, M (all types)

Structure of model `./graphene/graphene.qtdw`



arrow represents definition of positive bond direction

Construction `$ mp_create ./graphene_in.py`

Model output `./graphene/graphene.pdf`

Table 2: Harmonics — symmetry of SAMB

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
1	$\mathbb{Q}_0(A_{1g})$	A_{1g}	0	$Q,\ T$	-	-	1
2	$\mathbb{G}_1(A_{2g})$	A_{2g}	1	$G,\ M$	-	-	z
3	$\mathbb{Q}_3(B_{1u})$	B_{1u}	3	$Q,\ T$	-	-	$\frac{\sqrt{10}y(3x^2-y^2)}{4}$
4	$\mathbb{Q}_3(B_{2u})$	B_{2u}	3	$Q,\ T$	-	-	$\frac{\sqrt{10}x(x^2-3y^2)}{4}$
5	$\mathbb{Q}_{1,1}(E_{1u})$	E_{1u}	1	$Q,\ T$	-	1	x
6	$\mathbb{Q}_{1,2}(E_{1u})$					2	y

continued ...

rank, component (irrep.)

anisotropy

Table 3: dimension = 2 — basis of Hamiltonian matrix

#	orbital@atom(SL)	#	orbital@atom(SL)
0	$\lvert p_z\rangle$ @C(1)	1	$\lvert p_z\rangle$ @C(2)

Table 4: Atomic basis (orbital part only)

orbital	definition
$\lvert p_z\rangle$	z

definition of atomic basis

In the spinful case
x → (x,↑), (x,↓) etc.

MultiPie - Model (Example : Graphene)

Basis of site cluster

Table 7: 'C' (#1) site cluster (2c), -6m2

Wyckoff (site) and site symmetry

SL	position ($\boldsymbol{s}$)	mapping
1	[0.33333, 0.66667, 0.00000]	[1,2,3,10,11,12,16,17,18,19,20,21]
2	[0.66667, 0.33333, 0.00000]	[4,5,6,7,8,9,13,14,15,22,23,24]

fractional coordinate and mapping of symmetry op.

Basis of bond cluster (1st neighbor)

Table 8: 1-th 'C'-'C' [1] (#1) bond cluster (3a@3f), ND, $|\boldsymbol{v}|$= 0.57735 (cartesian)

Wyckoff (bond@site) and directionality, distance of bond

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[0.33333, 0.66667, 0.00000]	[0.50000, 0.00000, 0.00000]	[1,-4,-8,11,-13,16,20,-23]	(2,1)	(1,1)	[0,-1,0]
2	[-0.66667,-0.33333, 0.00000]	[0.00000, 0.50000, 0.00000]	[2,-5,-7,10,-14,17,19,-22]	(2,1)	(1,1)	[1,0,0]
3	[0.33333,-0.33333, 0.00000]	[0.50000, 0.50000, 0.00000]	[3,-6,-9,12,-15,18,21,-24]	(2,1)	(1,1)	[0,0,0]

fractional coordinate and mapping (sublattice, +trans.) $R = R_{\text{ket (tail)}} - R_{\text{bra (head)}}$

Basis of bond cluster (2nd neighbor)

Table 9: 2-th 'C'-'C' [1] (#2) bond cluster (6b@6l), ND, $|\boldsymbol{v}|$= 1.0 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[1.00000, 0.00000, 0.00000]	[0.83333, 0.66667, 0.00000]	[1,-11,16,-20]	(1,1)	(1,1)	[-1,0,0]
2	[0.00000, 1.00000, 0.00000]	[0.33333, 0.16667, 0.00000]	[2,-10,17,-19]	(1,1)	(1,1)	[0,-1,0]
3	[-1.00000,-1.00000, 0.00000]	[0.83333, 0.16667, 0.00000]	[3,-12,18,-21]	(1,1)	(1,1)	[1,1,0]
4	[-1.00000, 0.00000, 0.00000]	[0.16667, 0.33333, 0.00000]	[4,-8,13,-23]	(2,1)	(2,1)	[1,0,0]
5	[0.00000,-1.00000, 0.00000]	[0.66667, 0.83333, 0.00000]	[5,-7,14,-22]	(2,1)	(2,1)	[0,1,0]
6	[1.00000, 1.00000, 0.00000]	[0.16667, 0.83333, 0.00000]	[6,-9,15,-24]	(2,1)	(2,1)	[-1,-1,0]

MultiPie - Model (Example : Graphene)

Combined SAMB

total number of bases

— SAMB: 20 (all 20) —

no. of bases excluding duplications only with different bond length

- C : 'C' site-cluster name of site cluster
  - * bra: $\langle p_z|$
  - * ket: $|p_z\rangle$ basis of atomic SAMB
  - * wyckoff: 2c Wyckoff (site)

$\boxed{\text{z1}}$ $\mathbb{Q}_0^{(c)}(A_{1g}) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z5}}$ $\mathbb{Q}_3^{(c)}(B_{1u}) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_3^{(s)}(B_{1u})$

expression of combined SAMB
atomic x site-cluster

ID of basis

- C;C_001_1 : 'C'-'C' bond-cluster name of bond cluster (tail; head_neighbor_multiplicity)
  - * bra: $\langle p_z|$
  - * ket: $|p_z\rangle$ basis of atomic SAMB
  - * wyckoff: 3a@3f Wyckoff (bond@site)

$\boxed{\text{z2}}$ $\mathbb{Q}_0^{(c)}(A_{1g}) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z6}}$ $\mathbb{T}_3^{(c)}(B_{1u}) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{T}_3^{(b)}(B_{1u})$

$\boxed{\text{z9}}$ $\mathbb{T}_{1,1}^{(c)}(E_{1u}) = \dfrac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z10}}$ $\mathbb{T}_{1,2}^{(c)}(E_{1u}) = \dfrac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2}$

expression of combined SAMB
atomic x bond-cluster

matrix of atomic SAMB

- bra: $\langle p_z|$
- ket: $|p_z\rangle$

basis of atomic SAMB

$\boxed{\text{x1}}$ $\mathbb{Q}_0^{(a)}(A_{1g}) = \begin{bmatrix} 1 \end{bmatrix}$ 1 x 1 行列

vector of site and bond cluster

** Wyckoff: 2c Wyckoff (site)

$\boxed{\text{y1}}$ $\mathbb{Q}_0^{(s)}(A_{1g}) = \left[\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right]$ 2-site cluster

$\boxed{\text{y2}}$ $\mathbb{Q}_3^{(s)}(B_{1u}) = \left[\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right]$

** Wyckoff: 3a@3f Wyckoff (bond@site)

$\boxed{\text{y3}}$ $\mathbb{Q}_0^{(b)}(A_{1g}) = \left[\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right]$ 3-bond cluster

$\boxed{\text{y4}}$ $\mathbb{T}_3^{(b)}(B_{1u}) = \left[\frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{3}\right]$

$\boxed{\text{y5}}$ $\mathbb{T}_{1,1}^{(b)}(E_{1u}) = \left[0, -\frac{\sqrt{2}i}{2}, \frac{\sqrt{2}i}{2}\right]$

. . .

MultiPie - Physical Quantity (Example : Graphene)

Control file

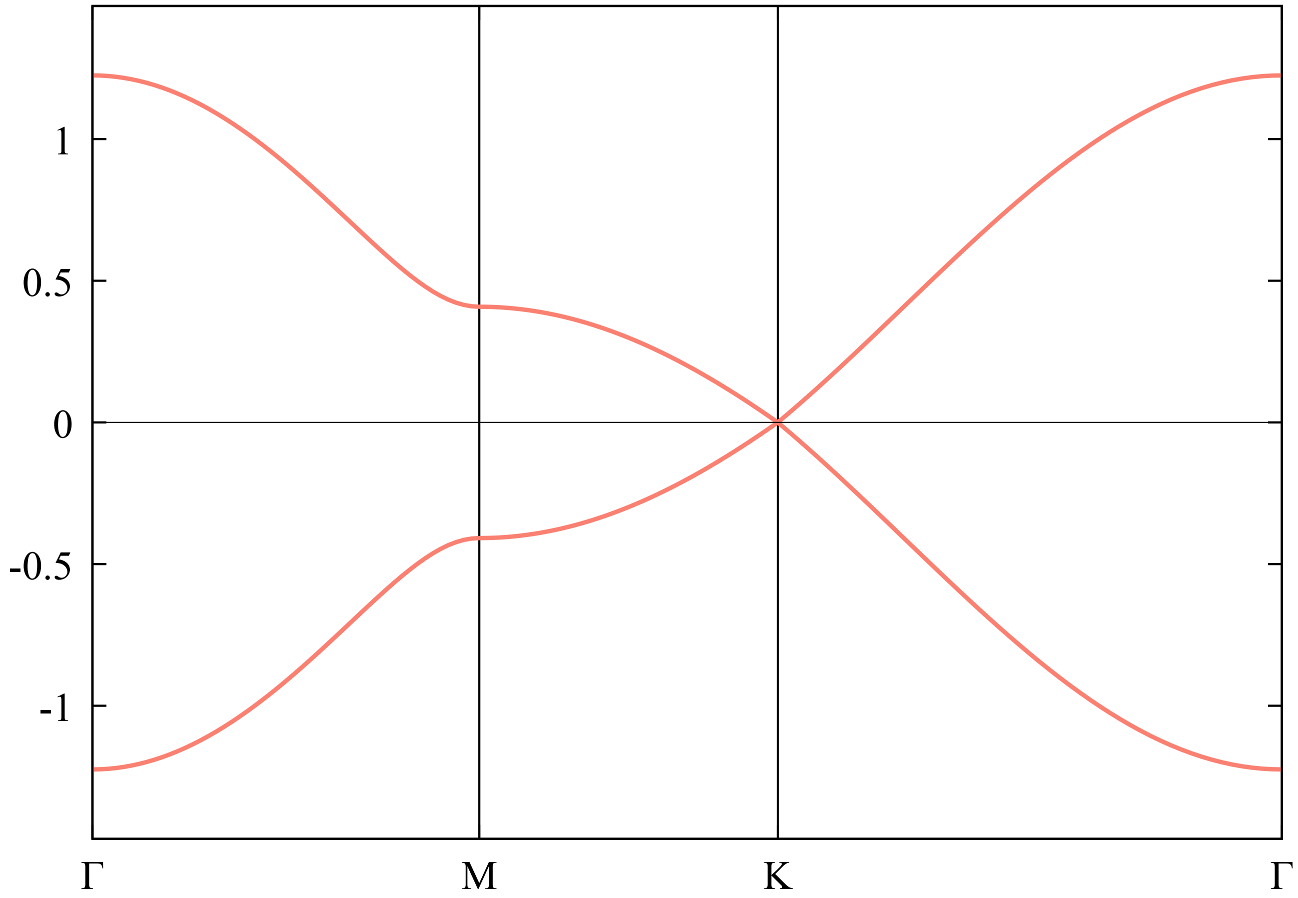
```
# graphene_ctrl.py
graphene_ctrl = {
    "samb": {
        "model": "graphene",                model name
        "parameter": {
            "z2": 1.0,                      ID and weight of SAMBs used
        },
        "samb_figure": True,                generate QtDraw file for SAMB ?
    },
    "output": {
        "dispersion": {
            "k_point": {"Γ": "[0,0,0]", "K": "[1/3,1/3,0]", "M": "[1/2,0,0]"},    definition of k point
            "k_path": "Γ-M-K-Γ",            high-symmetry line ( connected with "-", discontinuities with "|" )
        },
    },
}
```

Analyze & Output

(Now developing step by step)

```
$ mp_analyze ./graphene_ctrl.py
```

band dispersion `./graphene/output/graphene_dispersion.pdf`



MultiPie - Matrix Element (Example : Graphene)

Matrix-element file `./graphene/graphene_matrix.py`

```
graphene = {
    "model": "graphene",
    "source": "graphene.pkl",
    "date": "2026-07-12 10:23:48",
    "select": {     filter conditions for selecting basis
        "X": ["Q", "G", "M", "T"],
        "l": [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11],
        ...
    },
    "dimension": 2,   dimension of matrix
    "ket_site": {
        "pz@C(1)": [0.3333333333333333, 0.6666666666666666, 0.0],
        "pz@C(2)": [0.6666666666666666, 0.3333333333333333, 0.0]
    },
    "index": {("C", 1, 1): (0, 1), ("C", 2, 1): (1, 1)},   (atom, sublattice, orbital rank) → (top of block matrix, size of block matrix)
    "vector": {   correspondence between cluster and bond vector (primitive cell)
        "C": [[0.0, 0.0, 0.0]],
        "C;C_001_1": [[0.33333333, 0.66666666, 0.0], [-0.66666666, -0.33333333, 0.0], [0.33333333, -0.33333333, 0.0]],
        ...
   "cluster": {   correspondence between ID and cluster
        "z1": "C",
        "z2": "C;C_001_1",
        ...
    "matrix": {     ID : { (n1, n2, n3, m, n) : (matrix element, bond_no), ... }
        "z1": {(0, 0, 0, 0, 0): ("sqrt(2)/2", 0), (0, 0, 0, 1, 1): ("sqrt(2)/2", 0)},
        "z2": {(0, -1, 0, 1, 0): ("sqrt(6)/6", 1), (1, 0, 0, 1, 0): ("sqrt(6)/6", 2), ...,
       ...
    }
}
```

basis name = orbital@atom(sublattice) and frac. coord. (primitive cell)

$$\boldsymbol{k} = 2\pi(\kappa_1 \boldsymbol{b}_1 + \kappa_2 \boldsymbol{b}_2 + \kappa_3 \boldsymbol{b}_3)$$ fractional k vector (primitive cell)

matrix element in momentum space

$$[\mathbb{Z}_j(\boldsymbol{k})]_{m,n} = \sum c(n_1, n_2, n_3, m, n) e^{2\pi i \boldsymbol{\kappa}\cdot(\boldsymbol{R} - \boldsymbol{r}_m + \boldsymbol{r}_n)}$$

$$\boldsymbol{R} = \boldsymbol{R}_n - \boldsymbol{R}_m = n_1 \boldsymbol{a}_1 + n_2 \boldsymbol{a}_2 + n_3 \boldsymbol{a}_3$$

(primitive cell)

frac. coord.
in primitive unit cell

```
e.g.
  position = list( graphene["ket_site"].values() )
  r_m, r_n = position[m], position[n]
```

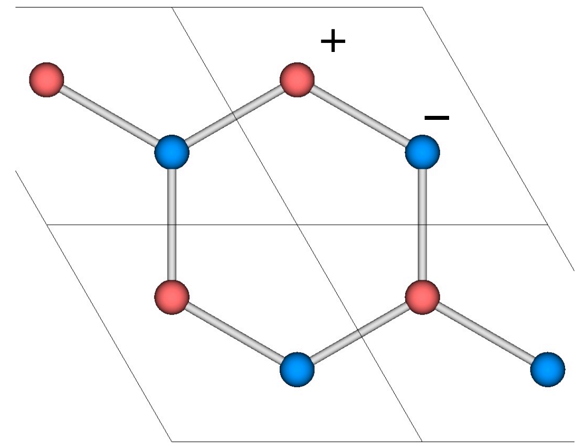
MultiPie - How to use basis (Example : Graphene)

Basis of non identity irreps.

mass term `samb/graphene_C.qtdw (y2)`

$$\boxed{\mathrm{z5}}\quad \mathbb{Q}_3^{(c)}(B_{1u}) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_3^{(s)}(B_{1u})$$

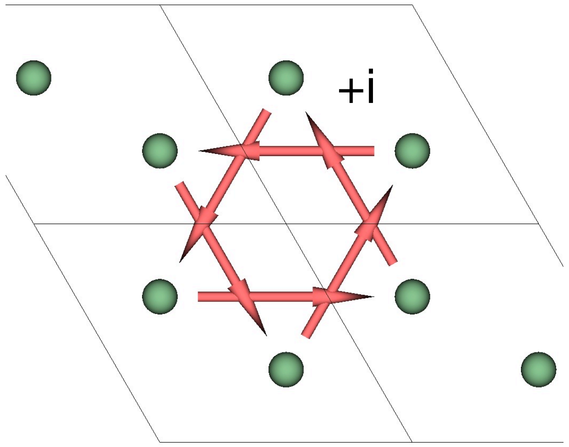
$$\boxed{\mathrm{y2}}\quad \mathbb{Q}_3^{(s)}(B_{1u}) = \left[\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right]$$



Haldane's magnetic flux `samb/graphene_C;C_002_1.qtdw (y10)`

$$\boxed{\mathrm{z4}}\quad \mathbb{M}_1^{(c)}(A_{2g}) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{M}_1^{(b)}(A_{2g})$$

$$\boxed{\mathrm{y10}}\quad \mathbb{M}_1^{(b)}(A_{2g}) = \left[\frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}\right]$$

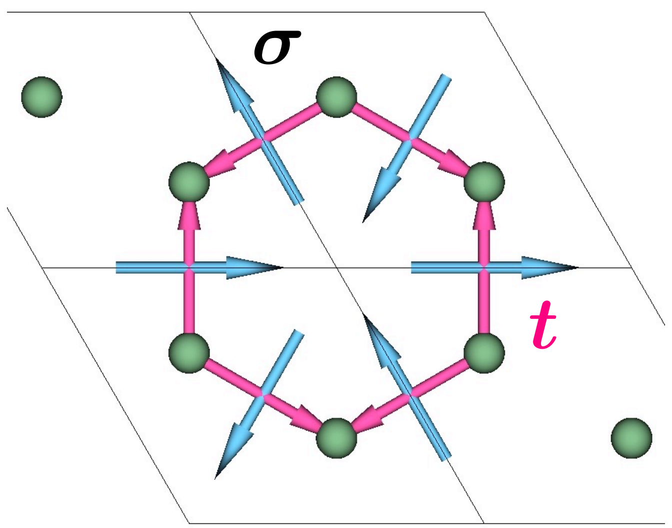


surface Rashba SOC if model is constructed with spinful = True, total number of bases becomes 80 (all 80)

$$\boxed{\mathrm{z12}}\quad \mathbb{Q}_1^{(1,-1;c)}(A_{2u}) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,-1;a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,-1;a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2}$$

$$\boxed{\mathrm{x3}}\quad \mathbb{M}_{1,1}^{(1,-1;a)}(E_{1g}) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & 0 \end{bmatrix} \qquad \boxed{\mathrm{x4}}\quad \mathbb{M}_{1,2}^{(1,-1;a)}(E_{1g}) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} \\ \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$$

$$\boxed{\mathrm{y5}}\quad \mathbb{T}_{1,1}^{(b)}(E_{1u}) = \left[0, -\frac{\sqrt{2}i}{2}, \frac{\sqrt{2}i}{2}\right] \qquad \boxed{\mathrm{y6}}\quad \mathbb{T}_{1,2}^{(b)}(E_{1u}) = \left[\frac{\sqrt{6}i}{3}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}\right]$$



Model Generation and Physical-Quantity Calculations

MultiPie - Workflow

1. Model construction

`from multipie import Group, MaterialModel`

`./input_file.py` — Python dict format `{"model"=name, ... }`

- atom position, bond, orbital
- selection of SAMB generation

`$ mp_create ./input_file.py` ← `MaterialModel` — `Group`

MaterialModel: Analyze input file and construct model with SAMB

Group: group-theoretical data, harmonics, SAMB data

output

`./name/name.tex`	
`name.pdf`	SAMB info. file (Summary)
`name.qtdw`	molecule, crystal structure (with QtDraw)
`name.pkl`	SAMB info. file (binary)

2. Plot band, physical quantities
(Now developing step by step)

`from multipie import ModelAnalyzer`

`./control.py` — Python dict 形式

- select SAMB
- weight or compute by projection
- CW or SymWannier input + calc., output

`$ mp_analyze ./control_file.py` ← `ModelAnalyzer`

ModelAnalyzer: from SAMB info., plot band dispersion, calculate physical quantities, ...

output

`./name/name_matrix.py`	matrix info. for selected SAMB
`/name_mp_hr.dat`	
`/output/...`	output of physical quantities
`...`	
`/samb/name_...`	visualization file for SAMB (with QtDraw)

MultiPie - Input File (mp_create)

Input file default values when omitted

```
default_model = {
    "model": "unknown",     model name : used as prefix for output folders/file names
    "group": "C1",          Crystallographic PG or SG name (Shoenflies symbol or SG no.)
    "cell": {},             unit-cell parameter (a, b, c, alpha, beta, gamma)
    "spinful": False,       spinful orbital ?
    "site": {},             rep. site in unit cell   "name": ( [frac. coord.], [orbital list] )
    "bond": [],             rep. bond in unit cell   ( "tail", "head", [neighbor list] )
    "SAMB_select": {  # select combined SAMB.        filter conditions in generating SAMB
        "X": ["Q", "G"],  # type, "Q/G/T/M", []=all.
        "l": [],  # rank, []=all.
        "Gamma": [],  # "IR"=identity, []=all, "..."/["...","..."]=specified irreps.
        "s": [],  # spin, 0/1, []=all.
    },
    "atomic_select": {  # select atomic SAMB.        filter conditions in generating atomic SAMB
        "X": [],  # type, "Q/G/T/M", []=all.
        "l": [],  # rank, []=all.
        "Gamma": [],  # "IR"=identity, []=all, "..."/["...","..."]=specified irreps.
        "s": [],  # spin, 0/1, []=all.
    },
    "site_select": {  # select site-cluster SAMB.     filter conditions in generating site-cluster SAMB
        "l": [],  # rank, []=all.
        "Gamma": [],  # "IR"=identity, []=all, "..."/["...","..."]=specified irreps.
    },
    "bond_select": {  # select bond-cluster SAMB.     filter conditions in generating bond-cluster SAMB
        "X": [],  # type, "Q/T/M", []=all.
        "l": [],  # rank, []=all.
        "Gamma": [],  # "IR"=identity, []=all, "..."/["...","..."]=specified irreps.
    },

    ...
```

MultiPie - Input File (mp_create)

```
  ...

  "toroidal_priority": False,      in generating SAMB, priority for toroidal multipoes ?
   "max_neighbor": 10,  # for DFT fitting, e.g. =200.      max. of neighbor in searching bond
   "search_cell_range": (-2, 3, -2, 3, -2, 3),             max. range of cell in searching bond
       # (a1, a2, a3) range. for DFT fitting, e.g.=(-10, 10, -10, 10, -10, 10).
   "qtdraw": {     option in generating QtDraw files
       "create": True,  # create QtDraw file ?     generate ?
       "mode": "standard",  # mode, "standard/detail".     standard or detail ?
       "view": None,  # [a,b,c], if None, default=[6,5,1].   default view
       #
       "rep_site": True,  # show representative site ?      highlight for rep. site ?
       "rep_bond": True,  # show representative bond ?      highlight for rep. bond ?
       #
       "max_neighbor": 5,  # max. neighbor to draw.          max. neighbor to plot bonds
       "cell_mode": None,  # "off/single/all", if None, single for SG, off for PG.  cell display mode
       "scale": None,  # default or magnification scale.     scale of site/bond size and width ?
       "site_radius": 0.07,  # base site radius.           default radius of site
       "bond_width": 0.02,  # base bond width.             default widht of bond
   },
   "pdf": {          option in generating PDF
       "create": True,  # create PDF file ?        generaete ?
       "common_samb": True,  # display common SAMB ?      display common bond SAMB ?
       "harmonics": True,  # display harmonics ?         display harmonics info. ?
       "max_neighbor": 5,  # max. neighbor to write (None: all).  max. of neighbor
   },
}
```

MultiPie - Control File (mp_analyze)

Control file default values when omitted

```
default_control = {
    "samb": {
        "model": "unknown",   model name or None
        "select": {      select condition of SAMB          S=site name, R=orbital rank, N=neighbor bond
            # "site": [("A", [1,2])],   list of S or ( S, [R] ) format
            # "bond": [[0,1,2]],      list of S1;S2, [N], ( S1;S2, [N] ), ( S1;S2, R1;R2, [N] ) format
            # "X": [],    list of SAMB type, Q/G/T/M ( [ ] = all types)
            # "l": [],    list of SAMB rank, 0, 1, ..., 11 ( [ ] = all ranks)
            # "Gamma": ["A1g"],    list of SAMB irrep. ( "IR" = identity irrep., [ ] = all irreps.)
            # "s": [],       list of SAMB internal rank, 0, 1 ( [ ] = 0 and 1)
        },
        "parameter": {
            # "z1": 1.0,  finite weight (float or sympy const.)  in case of empty dict. and  given wannier, compute by projection
        },                                                     In case of file name, read "./name/filename"
        "samb_figure": False,    output QtDraw files of SAMB ?
        "k_multipole": False,    create momentum rep. ?


    },
    "wannier": {  Wannier info. from DFT → Wannier computation
        "cw": None,   file name
    },
    "output": {       info. for computing physical quantities
        "dir": "output",      output folder
        "fourier": { "tb_gauge": True },      use atomic position for phase (TB gauge) ?
        "dos": False,    compute DOS ?
        "dispersion": {      for band dispersion
            "k_path": "",        high-symmetry line, connected with "-", discontinuities with "|" [for "", use default path, k_point is generated]
            "k_point": {"Γ": "[0,0,0]"},     definition of k-point (primitive fractional)
           "local": [],       expectation values of local operators (Sx, Sy, Sz, Lx, Ly, Lz, Qu, Qv, Qyz, Qxz, Qxy)
            "z": [],        SAMB ID to compute expectation value
        },
    },
}
```

MultiPie - コントロールファイル (mp_analyze)

Control file

default values when omitted

```
default_control = {
    "samb": {
        "model": "unknown",  model name or None
        "select": {  select condition of SAMB: S=site name, R=orbital rank, N=neighbor bond
            # "site": [("A", [1,2])],  list of S or (S,[R]) format
            # "bond": [[0,1,2]],  list of S1;S2, [N], (S1;S2,[N]), (S1;S2,R1;R2,[N]) format
            # "X": [],  list of SAMB type: Q,G,T,M ([]=all types)
            # "l": [],  list of SAMB rank: 0,1,...,11 ([]=all ranks)
            # "Gamma": ["A1g"],  list of SAMB irrep.: ("IR"=identity irrep., []=all irreps.)
            # "s": [],  list of SAMB internal rank: 0,1 ([]=0 and 1)
         },
        "parameter": {  finite weight (float/sympy) empty+wannier, compute by projection
            # "z1": 1.0,                   In case of file name, read "./name/filename"
         },
         "samb_figure": False,  output QtDraw files of SAMB ?
         "k_multipole": False,  create momentum rep. ?
         "NG_sum_rule": False,  adopt Nambu-Goldstone sum rule ?
    },
    "wannier": {  Use DFT→Wannier based info.
        "cw": None,  info file name not implemented yet
    },
    "output": {  info. for computing physical quantities
        "dir": "output",  output folder
        "fourier": { "tb_gauge": True },  use atomic position for phase (TB gauge) ?
        "dos": False,  compute DOS ?
        "dispersion": {  for band dispersion
            "power": None,  power for plot [None=1]
            "k_path": "",  high-symmetry path: connect by "-", discontinuity "|" [empty=default k_point generated)
            "k_point": {"Γ": "[0,0,0]"},  definition of k point (primitive,fractional)
           "local": [],  expectation values of local operators (Sx,Sy,Sz,Lx,Ly,Lz,Qu,Qv,Qyz,Qxz,Qxy)
            "z": [],  SAMB ID to compute expectation value not implemented yet
         },
    },
}
```

Group Class

Group - Overview

Initialization

Generate crystallographic point group, space group, magnetic PG, and magnetic SG

`Group(tag, with_opt=False)`

`tag`

	tag		example
point group	`"PG:no"`, `"Schoenflies"`	no = 1 - 47	`"PG:5"`, `"C3v"`
space group	`"SG:no"`, `"Schoenflies"`, `no`	no = 1 - 230	`"SG:152"`, `"D3^2"`, `152`
magnetic PG	`"MPG:HM_id"`	HM_id = PG.no.ID	`"MPG:12.3.8"`
magnetic SG	`"MSG:BNS_id"`	BNS_id = SG.no	`"MSG:152.3"`

`with_opt`

load additional data ? `group.opt[key]`

dict key	content
`harmonics_multipole`	multipolar harmonics (dipole, quadrupole, octupole)
`harmonics_irrep`	harmonics belonging to irrep. (irrep. → rank, no)
`representation_matrix`	representation matrix

Global info. `Group.global_info()`

dict key	content	dict key	content
`id_set`	list of tag classified by category	`harmonics`	max. of rank, name, definition, etc.
`tag`	unique tag → conventional tag	`response_tensor`	definition of tensor and multipole
`id`	conventional tag → unique tag	`root_cluster`	misc. of cluster SAMB
`character`	alias and compatibility relation		

Detail of API

https://cmt-mu.github.io/MultiPie/src/api_core/group.html

MaterialModel Class

MaterialModel - Overview

Initialization Create model from structure, symmetry, and atomic orbital info.

```
MaterialModel(topdir=None, verbose=False)
```

`topdir` top working directory (current dir when omitted)

`verbose` display log info. ?

Dict data `https://cmt-mu.github.io/MultiPie/src/api_core/material_model.html`

Property `MaterialModel.group`

Function

function name	content	function name	content
`X`	get atomic SAMB	`Y`	get cluster SAMB
`Z`	get combined SAMB	`analyze`	analyze input data
`cell_site_primitive`	get site position in primitive cell	`clear`	clear info.
`get_atomic_samb`	get all of atomic SAMB	`get_cluster_samb`	get all of cluster SAMB
`get_combined_samb`	get all of combined SAMB	`get_combined_samb_matrix`	get combined SAMB matrix
`get_hr`	get Hamiltonian matrix	`get_ket_site`	get ket and corresponding site
`get_multipole_expression`	get expression of multipole	`get_samb_matrix`	get full matrix info.
`ket`	list of ket basis in LaTeX format	`load`	load model file (binary)
`save`	save model info. in binary	`save_pdf`	save model info. in PDF
`save_samb_hr`	save SAMB matrix in hr format	`save_samb_matrix`	save SAMB matrix
`save_samb_qtdraw`	save QtDraw for SAMB	`save_view`	save structure file

Detail of API

`https://cmt-mu.github.io/MultiPie/src/api_core/material_model.html`

ModelAnalyzer Class

ModelAnalyzer - Overview

Initialization

Compute and output physical quantities from model info.

```
ModelAnalyzer(N1=50, N2=50, N3=50, topdir=None, verbose=False)
```

`N1, N2, N3` no. of divisions in 1st Brillouin zone (primitive cell)

`topdir` top working directory (current dir when omitted)

`verbose` display log info. ?

Property

property name	content	property name	content
HR	Hamiltonian matrix (real space)	**model**	MaterialModel object
name	model name	**output**	control (output related)
samb	control (samb related)	**wannier**	control (wannier related)

Function

`analyze(control)` read control file, compute physical quantities, and output

`control` control file name or dict.

`set_grid_size(N1, N2, N3)` set no. of divisions

`N1, N2, N3` no. of divisions in 1st Brillouin zone (primitive cell)

Detail of API

https://cmt-mu.github.io/MultiPie/src/api_core/model_analyzer.html